# 3 Yr. Degree/4 Yr. Honours 5th Semester Examination, 2025 (CCFUP)

## Subject : Computer Science

## Course : COMP5011 (MAJOR)

## (Data Communication and Networking)

**Time: 2 Hours** **Full Marks: 40**

*The figures in the margin indicate full marks.*
*Candidates are required to give their answers in their own words as far as practicable.*

1. Answer *any five* questions: 2×5=10

   (a) What is satellite communication?

   (b) What are the advantages of optical fiber over coaxial cable?

   (c) Define Computer Network.

   (d) What is Data Communication?

   (e) What do you mean by Star Topology?

   (f) What is DNS?

   (g) What are the primary functions of the transport layer?

   (h) What is the purpose of ICMP?

2. Answer *any two* questions: 5×2=10

   (a) Discuss 'RZ' and 'Manchester' digital to digital line coding technique with examples.

   (b) Explain one guided media and one unguided media of transmission.

   (c) Write the functions of session layer and data link layer of OSI model.

   (d) Discuss in detail about packet switching in computer network.

3. Answer *any two* questions: 10×2=20

   (a) (i) Discuss Hamming code with the help of example.

   (ii) What is TDM? 8+2

   (b) (i) What is Router? Discuss TCP in detail.

   (ii) What are the differences between static and dynamic routing algorithm? 2+6+2

   (c) (i) What do you mean by LAN? Discuss CSMA protocol.

   (ii) Write the function of a bridge. 2+6+2

(d) Write short notes on (*any two*): 5+5

(i) DNS protocol

(ii) Stop and wait ARQ

(iii) IP Address

(iv) Digital to analog modulation